Firm Heterogeneity and the Impact of Immigration: Evidence from German Establishments*

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Abstract

We document that large firms spend a higher share of their wage bill on immigrants than small firms, and that fixed costs of hiring immigrants explain this fact. We develop a model in which firms choose immigrant shares subject to fixed costs. Heterogeneity in immigrant shares affects welfare gains from immigration through two channels: native workers reallocate across firms, which affects immigrant–native competition; and prices fall more due to larger cost reductions, which are concentrated among large firms. Ignoring heterogeneity understates native welfare gains by 50%. We also study how trade interacts with immigrant-share heterogeneity to shape immigration gains.

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1 Introduction

Over the past two decades, the immigrant population in advanced economies has expanded by more than 80 percent, intensifying both academic and public debate over the consequences of immigration for native-born workers. A large body of research has studied this question, often implicitly assuming the existence of a representative firm. Yet firms differ along several dimensions such as size, export orientation, and workforce composition.

Using a rich dataset from Germany, we document a new dimension of firm heterogeneity: large employers are more immigrant-intensive than small employers. To the extent that immigration inflows reallocate resources across firms, differences in firms' immigrant shares may affect equilibrium adjustments in goods and labor markets. This paper asks whether and how heterogeneity in firms' immigrant shares affects our understanding of the effects of immigration on the welfare of native-born workers.

We characterize the relationship between employer size and immigrant intensity using establishment-level data from Germany between 2003 and 2011. We find a systematic relationship between employer size and immigrant intensity: the median establishment with more than 500 employees spends 5.6 percent of its wage bill on immigrants, whereas the median establishment with fewer than 10 employees spends zero. We also find evidence that fixed costs associated with hiring immigrant workers may drive this pattern. We exploit Germany's 2011 opening of its labor market to EU New Member States (NMS) as a natural experiment that lowered hiring costs of these immigrants. Using an event-study approach, we show that small establishments expanded their hiring of NMS workers more than large establishments, relative to comparable workers from non-EU countries, consistent with the presence of fixed costs in hiring migrants prior to 2011.[1]

Based on this evidence, we develop a model in which firms differ in their primitive productivity and their (endogenous) immigrant share to study the general equilibrium adjust-

[1] We also find evidence suggesting that workers from different countries may have comparative advantages in performing different tasks, providing a rationale for firms' willingness to pay these costs.

ment and welfare implications of an influx of immigrants. The model features tradable and non-tradable sectors, and consumers with CES preferences over goods within each sector. Each good is produced by a firm that can employ immigrant and native workers, who are imperfect substitutes. We model the immigrant hiring decision by closely following the intermediate input sourcing literature (Blaum et al., 2018). Firms may choose to hire immigrant labor, but doing so requires paying a fixed cost for each additional source country from which they recruit. Consequently, larger firms find it profitable to recruit immigrants from more countries and to spend a larger share of their wage bill on immigrants. To fully capture the observed relationship between firm size and immigrant share, the model allows for firm-specific fixed costs of hiring immigrants, which are drawn from a joint distribution with innate productivity.

We use a simplified model to show analytically that the welfare predictions of a model that ignores the relationship between firm size and immigrant share are biased. We do so by comparing welfare gains in a model with and without heterogeneity in immigrant intensities (but with heterogeneity in productivity). First, the competition between immigrants and natives in the labor market depends on the aggregate elasticity of substitution. With homogeneous immigrant shares across firms, the aggregate elasticity of substitution coincides with the within-firm elasticity. With heterogeneous firms, however, the aggregate elasticity also reflects substitution across firms (or goods), which governs changes in firms' scale of production. When the (within-firm) substitution effect is stronger than the scale effect, immigrants crowd out natives at immigrant-intensive firms, who are reallocated toward native-intensive firms. This reallocation allows natives and immigrants to specialize in producing different goods, making the two groups less substitutable in the labor market. This, in turn, mitigates the wage pressure from immigration relative to the homogeneous case. Second, immigration generates a larger decline in aggregate prices because firms' marginal costs are more elastic with respect to immigrant wages, and these cost reductions are disproportionately concentrated among the most productive firms. Overall, the *sign of the bias* depends on whether the elasticity of substitution between immigrants and natives is larger or smaller than the elasticity of demand, while

the *magnitude of the bias* depends on the joint distribution of firm productivity and immigrant-hiring costs.

We also show analytically that immigration gains in an open economy can be larger or lower than in a closed economy due to firm heterogeneity in immigrant share. In an open economy, the larger market size amplifies the scale effect at immigrant-intensive firms, reducing the displacement of native coworkers and weakening reallocation toward native-intensive firms. At the same time, the accompanying increase in profits induces firms to pay the fixed cost to source immigrants from more countries, increasing substitution away from natives and thus reinforcing reallocation toward native-intensive firms.

We estimate the elasticity of demand from the average firms' markups, following Oberfield and Raval (2014). The (within-firm) substitution between immigrants and natives is structurally estimated using the firm's first-order condition with respect to immigrant and native labor. We regress the firm-level relative wage bill between immigrants and natives on relative employment. We extend Ottaviano and Peri (2012)'s approach to allow for heterogeneity in workers' abilities, as suggested by our model, and using a model-based firm-level instrument. At a 1% confidence level, our estimates suggest that the elasticity of substitution between immigrants and natives within firms exceeds the elasticity of demand. Thus, the observed heterogeneity in immigrant share is expected to generate larger welfare gains from immigration. We calibrate other parameters jointly to match key targeted micro- and macro-level moments in Germany between 2003 and 2011. Importantly, the observed relationship of immigrant share and firm size, our novel fact, disciplines the parameters of the joint distribution of productivities and fixed costs.

We verify that the calibrated model properly captures the mechanisms through which firm heterogeneity in immigrant share affects welfare. Our analytical results show that the heterogeneity in immigrant shares leads to heterogeneity in the elasticity of sales and the elasticity of the immigrant-to-native wage bill ratio to immigrant labor inflows. Therefore, we compare the predicted changes in firms' sales and the immigrant-to-native wage bill ratio due to an exogenous immigrant inflow between the model and the data.

To that end, we regress firm revenues and the relative wage bill ratio on the share of immigrants in the local labor market and its interaction with firm size. We use these reduced-form estimates to compute the effect of a 1% increase in the share of immigrants in the labor market for firms in different size deciles and find that the model does a good job of matching its data counterparts.

We use the calibrated model to measure the real wage (or welfare) effects of a 20% increase in the total number of immigrants, motivated by the inflow experienced by Germany between 2011 and 2017. We find that native workers in both sectors benefit from immigration despite the increased competition in the labor market, as the cost of the consumption basket falls relative to their wages. Production increases for both sectors, but more so in the tradable sector, where firms are more intensive in immigrant labor. Firms in the top size decile of the tradable sector, being the most immigrant-intensive, absorb a large share of incoming immigrants, leading to the reallocation of many native workers to other firms. Given their size, large firms in the non-tradable sector absorb a significant portion of these native workers. This reallocation between firms explains most of the cross-sector reallocation of native workers. Natives also reallocate within sectors toward less immigrant-intensive firms. Overall, welfare gains of native-born workers and firm owners are 0.11% and 1.31%, respectively. In monetary terms, these gains amount to $1.8 billion for native workers and $20.8 billion for firm owners.[2]

We find that the quantitative relevance of both analytical results is large in the aggregate. First, the version of our model that removes this within-industry heterogeneity in immigrant shares, while still allowing immigrant shares to vary across industries and Hicks-neutral productivity to vary across firms, predicts welfare gains roughly 50% lower. Second, in a counterfactual trade-autarky economy, welfare gains from immigration are approximately 10% larger. Of this trade-dampening effect, 60% is accounted for by within-sector heterogeneity in firms' migrant shares.[3]

[2]These effects are relatively large. For reference, estimates of the U.S. welfare gains from China's rise in world trade range from 0.03% to 0.2% (Hsieh and Ossa (2016), Caliendo et al. (2019)).

[3]The remaining 40% (0.004 pp) reflects mechanisms that arise even under firm homogeneity (i.e., terms of trade effects) and is remarkably similar to previous findings (Caliendo et al., 2019).

Our paper contributes to the literature on the economic effects of immigration on the receiving economy (e.g., Card (2001), Card (2009), Ottaviano and Peri (2012)). A stream of this literature uses general equilibrium (GE) models that jointly determine wages and goods prices to study the impact of immigration on the levels of real wages (e.g., di Giovanni et al. (2015), Desmet et al. (2018), Burstein et al. (2020), Caliendo et al. (2021), Khanna and Morales (2025), Morales (2025)). These GE models, unlike ours, restrict immigrant-share heterogeneity to broader units of production (e.g., industries) and do not allow it to vary across firms. By introducing heterogeneous firms that endogenously choose their immigrant intensities due to fixed costs to hire immigrants, we make two contributions. First, our results suggest that labor reallocation across firms within sectors is an important adjustment channel for aggregate production and labor markets. Second, the aggregate elasticity of substitution does not inform welfare effects when firm heterogeneity in immigrant share arises from fixed costs.[4]

We also contribute to the stream of the empirical literature studying the effect of immigration on firms (e.g., Dustmann and Glitz (2015), Mitaritonna et al. (2017), Arellano-Bover and San (2023), Amior and Stuhler (2024), Gyetvay and Keita (2024), Mahajan et al. (2025)) by documenting new facts on the relationship between firm size and immigration and assessing the aggregate general equilibrium effects of immigration.[5]

Our work contributes to the international trade literature studying the role of trade in shaping the effects of immigration, which dates back to Rybczynski (1955) and Samuelson (1948). Recent work quantifies the relevance of trade using modern quantitative models that abstract from firm heterogeneity (e.g., Caliendo et al. (2021) and Brinatti and Guo (2024)). We contribute to this literature by studying whether trade amplifies or dampens immigration gains due to a new mechanism, firm heterogeneity in migrant shares.[6]

[4] Mehra and Shen (2022) and Mandelman et al. (2025) also propose models that include hiring costs to hire immigrants in the U.S. Their models are calibrated using mostly aggregate data.

[5] Other notable contributions include Kerr et al. (2015), Beerli et al. (2021), Egger et al. (2022), Card et al. (2023), Mahajan (2024), and Orefice and Peri (2024).

[6] Our result that the aggregate elasticity is not a sufficient statistic for the effects of immigration on native welfare relates to the stream of the trade literature studying whether firm heterogeneity matters for the welfare effects of trade (Arkolakis et al., 2012; Melitz and Redding, 2015).

This paper also contributes to the literature on the relationship between micro and macro elasticities. Existing work shows that firm heterogeneity in factor shares leads to differences between the micro and macro elasticities of substitution between factors (e.g., Satō (1975), Oberfield and Raval (2021), Lashkari et al. (2024)). This work tends to focus on aggregate variables such as the labor share rather than the implications for welfare. We link firm heterogeneity in factor shares and micro elasticities to welfare, and explain that the macro elasticity is not sufficient for welfare analysis in our fixed-cost model.

2 Data

We use a detailed, employer-employee matched dataset from Germany provided by the Institute for Employment Research (IAB). The main data source is the Longitudinal Establishment Panel (LIAB), which includes records for a large sample of establishments over the period 2003-2011 (Heining et al., 2016).[7] The dataset contains full employment trajectories for each employee who worked at least one day for one of the establishments in the sample during the period. It also includes employee information on citizenship, occupation, education, and daily wage. While our ideal measure for immigration status would be country of birth, the data reports citizenship, where countries are grouped into ten regions. We classify as immigrants those who report non-German citizenship in at least two periods. Appendix **??** discusses the details and robustness of this definition.

On the establishment side, the dataset contains information on industry, location, and financials such as revenues, material use, and export share, among others (Institut für Arbeitsmarkt- und Berufsforschung (IAB), 2013). It is important to note that the data is at the establishment level, and it is not possible to link multiple establishments to a single firm. Throughout the paper, we use establishment and firm interchangeably.[8] Also, we apply survey weights to get representative aggregates whenever necessary.

[7] The data basis of this paper is the Longitudinal Model (version 1993–2014) of the Linked Employer-Employee Data from the IAB. The data were accessed on-site at the Research Data Centre of the Federal Employment Agency at the Institute for Employment Research (FDZ) and remotely.

[8] Our main facts are robust to looking solely at single-establishment firms.

Finally, we use the Sample of Integrated Employer–Employee Data (SIEED, Berge et al. (2020)) in Section 3.3.2 to provide evidence on the presence of fixed costs, as it includes post-2011 information and workers' country of citizenship.[9] This administrative data is a 1.5% random sample of the universe of German establishments, where we also access the entire labor market biographies of employees who worked at least one day in these establishments.

3 Firm Heterogeneity in Immigrant Share

We begin our analysis by establishing a key fact: larger establishments are more intensive on immigrant labor than smaller establishments. To assess the implications of this heterogeneity for the welfare effects of immigration on native workers, we must understand the microfoundations that determine immigrant employment shares. To do so, we begin by discarding natural potential explanations that are inconsistent with salient features of the data. Second, we show that fixed costs associated with hiring immigrant workers likely constitute the primary explanatory factor of the observed systematic relationship between employer size and immigrant intensity.[10]

3.1 Fact: Immigrant intensity increases with employer size

To illustrate the relationship between employer size and immigrant share, we classify the establishments in our sample into ten bins based on their total reported employment. For each bin, we plot the median share of immigrant labor in the establishment wage bill to capture the firm-level intensity of immigrants.[11] Figure 1 shows a systematic relationship between employer size and immigrant intensity. The median establishment among firms with more than 500 employees spends 5.6% of their wage bill on immigrants, while the median establishment with less than 10 employees spends zero.

[9] We do not use SIEED for the rest of the analysis because it does not have firms' financial information.

[10] Appendix **??** presents summary statistics on the sample of establishments and the distribution of immigrants across sectors and origin regions.

[11] Our findings are robust to alternative bin definitions (e.g., deciles of revenues, or wage bill) and to using statistics other than the median, such as calculating the immigrant share for each bin, or running a regression of immigrant share on establishment size. We prefer using the median due to outliers.

3.2 Ruling out potential mechanisms

We consider potential underlying mechanisms for the documented heterogeneity. One possibility is that large employers concentrate in industries that use immigrant-supplied skills more intensively or in locations with larger immigrant networks. However, the pattern remains strong after controlling for three-digit industry fixed effects and local labor market fixed effects (Appendix Figure **??**).

Another explanation is that immigrants might reduce trade costs between origin and destination countries (Ottaviano et al. (2018), Bonadio (2023)), which would be more relevant for exporters because they are typically larger employers. Yet, exporters and non-exporters display similar patterns (Appendix Figure **??**), suggesting that reductions in employers' trade costs are unlikely to be the main explanation.

A third explanation is due to immigrant-biased technological differences across establishments of different sizes. This explanation implies a common *elasticity* of immigrant intensity to immigration across establishments, which is counterfactual (see Section 5.5).

Appendix **??** considers additional channels. For instance, we show that our key fact is not driven by immigrant skills and large establishments being more skill-intensive, as we find the pattern holds within skill groups. Similarly, we show our pattern is not explained by the establishment being a multinational or part of a multi-establishment firm.

3.3 Fixed costs to hire immigrants

3.3.1 Institutional aspects and descriptive evidence of fixed costs

During our period of analysis, most non-EU nationals were allowed to migrate to Germany only after securing a job offer. This arrangement, similar to the U.S. system, places the burden of navigating administrative procedures, as well as identifying and screening candidates, on the sponsoring employer (see Appendix Section **??**).

In addition to these policy-induced fixed costs, firms may incur costs required to build

a recruitment channel from specific source countries. For example, firms may need to develop internal capacity to evaluate foreign credentials or learn how to access networks of suitable workers from a given origin country. This is consistent with evidence from the OECD and the German Chamber of Commerce and Industry (DIHK), who ran an employer survey in 2010 asking for the reasons why firms with unfilled vacancies did not attempt to hire workers from abroad (OECD, 2013). The top three reasons for this trend are the lack of German language skills of candidates, unclear and complex administrative procedures, and difficulties contacting candidates abroad.

The survey also shows that these barriers disproportionately impact small and medium enterprises (SMEs). One reason is that SMEs have fewer resources than large firms to overcome these barriers. For instance, larger firms are better positioned to afford specialized HR services that navigate the complexities of the immigration system and facilitate contacting candidates abroad. Moreover, the German Employment Agency verifies that the employer's petition is legitimate and that the working conditions offered to the foreign worker are not below those offered to German employees in the same occupation. According to the report, these checks tend to be more severe when the employer is not well-known, as tends to be the case for SMEs. The difficulty of SMEs to hire immigrants has even been the subject of public policy debate.

We complement this anecdotal evidence with direct evidence from the data, including the substantial share of establishments that do not hire immigrants and the lumpiness of the hiring process (see Appendix Section **??**).

3.3.2 Causal evidence of fixed costs to hire immigrants before 2011

This anecdotal and cross-sectional evidence is suggestive of fixed costs, but it does not isolate their effect from other unobserved factors that may influence immigrant hiring decisions across employers of different sizes. We make progress in this direction by using the 2011 incorporation of the New Member States (NMS) into the EU, which granted

citizens from 10 countries unrestricted access to the German labor market.[12] This policy change provides a natural experiment that lowered fixed and variable hiring costs for NMS immigrants relative to non-EU nationals, who continued to face the same restrictions. Consistent with the existence of fixed costs to hire these immigrants before 2011, we find that smaller establishments exhibited a stronger increase in NMS hiring than larger establishments.

To use time and cross-section variation introduced by this policy change, we use SIEED data because it includes several years after the policy change and citizenship country information. We estimate the effect of the policy on small and large firms' hiring of NMS immigrants using the following event-study framework:[13]

$$\frac{Emp_{o,j,t}}{Emp_{j,2010}} = \sum_{\tau \neq 2010} \left(\theta_\tau \times \mathbb{1}_j + \beta_\tau^L\right) \times \mathbb{1}_o \times \mathbb{1}_t + FE_{o,j} + FE_{j,t} + \epsilon_{o,j,t} \quad (1)$$

where $\mathbb{1}_j \equiv \mathbb{1}(j \in SME)$, $\mathbb{1}_o \equiv \mathbb{1}(o = NMS)$, and $\mathbb{1}_t \equiv \mathbb{1}(t = \tau)$. $Emp_{o,j,t}$ is the employment of nationals from group o at firm j in year t, and $Emp_{j,2010}$ is firm j's total employment in 2010, the year before the policy change. We consider two country groups: NMS members, which are the treated group, and European countries that do not yet belong to the European Union, which are the control group. SME is the group of small and medium establishments with fewer than 200 employees in 2010. $FE_{o,j}$ are fixed effects at the country group-establishment level that control for pre-existing differences between immigrant groups and establishments, such as a tendency of establishment j to hire NMS nationals. $FE_{j,t}$ are fixed effects at the establishment-year level that account in a flexible way for the effect of demand and cost shocks faced by the establishment that affect its hiring decisions. $\epsilon_{o,j,t}$ is the error term, which we cluster at the establishment level.

The coefficients β_τ^L measure the differences in the outcome variable for large firms between year t and 2010, our baseline year, for the treated group relative to the control group.

[12] Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia, and Slovenia joined the EU in 2004 but gained access to the labor market in Germany in 2011.

[13] Appendix Figure **??** shows the estimates of the average effect pooling together all firms, β_τ.

Similarly, the coefficients $\beta_\tau^{SME} \equiv \theta_\tau + \beta_\tau^L$ measure the effects for small firms. For example, a positive β_τ^L for $\tau > 2010$ indicates that, on average, large establishments increased the employment of NMS nationals relative to the control group after 2010. The regression restricts the sample to firms that did not hire either treated or control group immigrants in 2005 to transparently show the effect of the policy on the extensive margin. Results are robust to not imposing this restriction, as shown in Appendix Table **??**. The identifying assumption is that, conditional on the controls, the treated and control groups would have followed similar trends in the absence of the policy change. We assess the plausibility of this assumption by formally testing whether β_τ^{SME} and β_τ^L are zero before 2010. Failing to reject this hypothesis would suggest that the hiring trends for nationals from the treated and the control group were parallel before 2010.

Figure 2 plots the estimated coefficients β_τ^L and β_τ^{SME}, suggesting that establishments began to increase their hiring of NMS nationals relative to the control group only after the restrictions on NMS nationals were lifted. More importantly, it suggests that smaller establishments increased the hiring of NMS nationals relative to large establishments.[14] We interpret the findings of this section as providing evidence of policy-induced fixed costs to hire immigrants that were in place until 2011, our period of analysis.

Robustness exercises: Given that lower-ability workers tend to work at lower productivity firms (Card et al., 2013), our estimates could be biased upward if the post-2011 inflow of immigrants had lower abilities compared to those who arrived pre-2011. However, our data suggests that, if anything, the new NMS migrants are positively selected in the post-period. For instance, the wages paid to NMS increase by 9.8% between the pre- and post-period, while they increase by 8.2% for immigrants in the control group.

Appendix Section **??** shows that the differential response of small and large establishments is robust to alternative specifications. For conciseness, we report the difference-in-difference estimates analogous to equation 1. Table **??** considers alternative outcome

[14]Moreover, column 1 of Appendix Table **??** shows that this decreasing effect on employer size persists even when the SME group is further disaggregated.

variables such as new hires, new entrants in the labor market employed by the establishment, and the immigrant share. Table **??** shows that the results are robust to using the full sample of establishments, rather than restricting to those that did not employ treated or control group immigrants in 2005, and to restricting the sample to establishments with more than 10 employees. Finally, Table **??** shows that results are robust to including an alternative fixed effects, such as origin-time fixed effects.

3.4 Additional evidence guiding modeling assumptions

For employers to optimally hire both immigrants and natives despite fixed costs, such workers must differ in the workplace. If they were perfect substitutes, firms would hire either only native-born or immigrant-born workers. Appendix **??** shows that immigrants from different countries have different bundles of characteristics — such as age, gender, and education — and specialize in different occupations than natives. We also provide evidence consistent with this specialization being related to origin-specific comparative advantage. Based on this evidence, our model will treat immigrants and natives, as well as immigrants from different origin countries, as imperfect substitutes in production.

4 The Model

This section studies how firm heterogeneity in immigrant share shapes the economy's adjustment to immigration inflows and the effects on native workers' welfare. We develop a model informed by the evidence from previous sections. Firms with heterogeneous productivities seek to hire immigrant labor because immigrants are imperfect substitutes for native workers. However, doing so requires two types of fixed costs: an initial cost to begin hiring immigrants and an additional cost for each new country they source immigrants from. This cost structure implies that larger, more productive firms spend a larger share of their wage bill on immigrants (Figure 1), are more likely to hire immigrants (Figure **??**), and more likely to recruit immigrants from more countries (Figure **??**) compared to smaller firms. We use the model to study the role of firm heterogeneity

in immigrant share in the effects of an immigrant labor supply shock. We focus on immigrant *supply* shocks to contribute to the large immigration literature studying the effects of immigrant inflows on the receiving economy.[15] We focus on the main components of the model and relegate derivations to Appendix **??**.

4.1 Setup

Environment: The world comprises two regions, Germany and the rest of the world (RoW), indexed by g and r respectively. The latter being composed of a continuum of countries indexed by o. In Germany, there is a tradable and a non-tradable sector indexed by $k = \{T, NT\}$, each populated by N_k firms indexed by j. Good markets are monopolistically competitive and labor markets are perfectly competitive.

Consumption: Consumers in Germany have Cobb-Douglas preferences for goods from the two sectors and CES preferences over varieties z within a k:

$$U = (Y_T)^{\alpha}(Y_{NT})^{1-\alpha} \quad , \quad Y_k = \left(\int_{J_k} (y_k(z))^{\frac{\sigma_d - 1}{\sigma_d}} dz \right)^{\frac{\sigma_d}{\sigma_d - 1}} \tag{2}$$

Y_k is the consumption of goods from k, α is the expenditure share of the tradable sector, J_k represents a fixed set of varieties available in k, and $\sigma_d > 1$ is the domestic elasticity of demand. Consumers are either firm owners, whose income is profits, or workers who earn wages. They choose $y_k(z)$ to maximize U subject to their income.

Production: Each firm produces a variety by combining native and foreign effective units of labor (d_j and x_j, respectively) in a CES manner as shown in equation 3. This modeling assumption is consistent with our evidence and follows a long tradition in the immigration literature, which understands immigrants and native-born workers as having comparative advantages in different tasks (Peri and Sparber, 2009, 2011).

$$y_j = \psi_j \left(\beta_k d_j^{\frac{\epsilon - 1}{\epsilon}} + (1 - \beta_k) x_j^{\frac{\epsilon - 1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon - 1}} \tag{3}$$

[15] In Section 6, however, we also study the effects of policies that reduce fixed costs to hire immigrants.

where β_k is a sector-specific distributional parameter, ϵ is the elasticity of substitution between native and immigrant workers within the firm, and ψ_j is a firm-specific productivity draw. The marginal cost of production is given by equation 4:

$$u_j = \left(\beta_k^{\epsilon} w_{d,k}^{1-\epsilon} + (1-\beta_k)^{\epsilon} W_{x,j}^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}} \tag{4}$$

where $w_{d,k}$ is the wage per effective unit of native labor and $W_{x,j}$ is the firm-specific wage per effective unit of immigrant labor, respectively. The share of native workers in the firms' wage bill is:

$$s_j = \frac{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon}}{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon} + (1-\beta_k)^{\epsilon} W_{x,j}^{1-\epsilon}} = \frac{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon}}{u_j^{1-\epsilon}} \tag{5}$$

If the wage per effective unit of immigrant labor, $W_{x,j}$, were identical across firms in sector k, all firms would choose the same immigrant and native shares. However, Section 3 shows that large firms are more immigrant-intensive than small firms. To capture this fact, we need a theory based on our evidence on why firms have different s_j.

Environment to Recruit Immigrants: To theorize on the firm choice of its immigrant share that accommodates our facts and remains tractable in general equilibrium, we follow Blaum et al. (2018)'s model for intermediate input sourcing. We assume that the immigrant labor input, x_j, is a composite of labor from different origin countries:[16]

$$x_j = \left(\int_{\Sigma_j} \delta_o \, x_{j,o}^{\frac{\kappa-1}{\kappa}} \, d_o\right)^{\frac{\kappa}{\kappa-1}} \tag{6}$$

where δ_o is the productivity of workers from country o, $\kappa > 1$ is the elasticity of substitution across foreign origin countries, and Σ_j is the hiring strategy of the firm given by the countries o where the firm hires immigrants from. Given that immigrants from different origins are imperfect substitutes, every additional origin country the firm hires immigrants from will have a positive impact on productivity and lower firm j's effective immigrant unit cost $W_{x,j}$.

[16] Appendix Section **??** suggests that immigrants from different countries may serve as different inputs in production due to differences in comparative advantage.

Following the evidence presented in Section 3.3, we assume firms must pay a fixed cost f_k^{imm} to begin hiring immigrants from abroad and a firm-specific fixed cost f_j for each additional origin country it wants to hire from. For example, if the firm hires immigrants from two origins, it spends $w_{d,k}(f_k^{imm} + 2f_j)$ in hiring costs. One interpretation is that the fixed cost f_k^{imm} captures the costs of setting up a legal department or training HR staff in order to start hiring immigrants. The cost f_j captures the learning cost that is country-specific, such as spending resources to understand foreign education credentials and labor experience necessary to screen workers.

We make two assumptions to maintain tractability in general equilibrium. First, δ_o in equation 6 is common across firms. Firms first source from the foreign country with the largest δ_o and move down the ladder as they source from more countries. This assumption reduces the dimensionality of the sourcing problem to choosing the mass of countries $n \in [0, 1)$ to hire from. Second, we assume δ_o is a random variable distributed Pareto with shape parameter ξ and scale parameter $\bar{\delta}$. These assumptions allow us to get a closed-form expression for the wage index of immigrants as in equation 7:

$$W_{x,j} = w_{x,k}\, \bar{z}\, n_j^{-\iota} \tag{7}$$

where $w_{x,k}$ is equilibrium wage per effective unit immigrants in Germany.[17] The parameters $\bar{z} > 0$ and $\iota > 0$ are determined by $\bar{\delta}$, ξ, and κ and can be interpreted as the average productivity of immigrants and the elasticity of the immigrant unit cost to expanding the mass of sourcing countries. Intuitively, imperfect substitution of immigrants generates productivity gains from hiring immigrants from additional origins. This reduces the wage index of immigrants and the unit cost of production.

Distributional assumptions of ψ_j and f_j: We assume that productivity ψ_j and hiring cost f_j are jointly drawn from a multivariate sector-specific lognormal distribution

[17] In equilibrium, immigrants earn the same wage per efficient unit regardless of origin. This equilibrium result enables the closed-form solution in equation 7, which allows us to keep tractability in general equilibrium while letting efficient wages for both immigrants and natives be determined endogenously.

with mean $[\mu_k^{\psi}, \mu_k^{f}]$, dispersion $[\sigma_k^{\psi}, \sigma_k^{f}]$, and covariance $\sigma_k^{\psi,f}$.[18]

Pricing Decision: Given CES preferences, the optimal price p_j is a constant markup over the marginal cost.

Optimal native share: An advantage of this setup is that we can write u_j, p_j, and n_j as functions of the key empirical object s_j as follows:

$$p_j = \frac{\sigma_d}{\sigma_d - 1} \frac{1}{\psi_j} \underbrace{\beta_k^{\frac{\epsilon}{1-\epsilon}} \; w_{d,k} \; s_j^{\frac{1}{\epsilon-1}}}_{u_j} \tag{8}$$

$$n(s_j) = \bar{\chi}_k \left(\frac{1}{s_j} - 1 \right)^{\frac{1}{\iota(\epsilon-1)}} \tag{9}$$

where $\bar{\chi}_k \equiv \left((\frac{1-\beta_k}{\beta_k})^{\epsilon} \; \bar{z}^{1-\epsilon} \right)^{-\frac{1}{\iota(\epsilon-1)}} (\frac{w_{x,k}}{w_{d,k}})^{\iota}$. Equation 9 follows from equations 5 and 7.

Firms maximize their profits by choosing the optimal native share s_j:

$$\max_{s_j} \; \Pi_j = \underbrace{\left(p_j(s_j) - \frac{u_j(s_j)}{\psi_j} \right) y_j}_{\text{variable profits}} - \underbrace{n_j(s_j) f_j w_{d,k} - w_{d,k} f_k^{imm} \mathbb{1}(n_j(s_j) > 0)}_{\text{Sourcing cost}} \tag{10}$$

How do firms choose their optimal native share? They face a trade-off between the drop in the marginal cost of production induced by the complementarity of hiring from an additional country and the fixed cost to source from that additional country. Given their scale of production, larger firms earn higher profits and can afford to pay f_j more times than small firms. Thus, larger firms hire immigrants from more countries than small firms, and they become more immigrant-intensive.

An important takeaway of the role of s_j in understanding the effects of immigration is the following. Firms benefit from an immigration inflow because the relative wage of immigrants drops, and so does the unit cost of production. The size of the drop in the unit cost of production is firm-specific, and it depends on the firm's native share. In

[18] The stricter verification procedure faced by small firms mentioned in Section 3.3 serves as a example of a negative correlation between f_j and ψ_j.

other words, the native share acts as a firm exposure to a common immigration shock and becomes the key empirical object to learn about how much each firm benefits from immigration.

Export Decision: Firms in the tradable sector can export their goods by paying a fixed cost f_x, as in Melitz (2003). The firm exports if the variable profits from export sales are larger than f_x. Under the assumption that consumers in RoW have CES preferences over tradable varieties with elasticity of demand σ_x, the optimal export price p_j^x is a constant markup over marginal cost, which now includes an iceberg cost $\tau > 1$: $p_j^x = \frac{\sigma_x}{\sigma_x - 1} \frac{u_j \tau}{\psi_j}$

Rest of the world: Given our focus on the German economy, we simplify the modeling of the RoW. We assume it has a single tradable sector, firms are equally productive, and use only native labor to produce with a constant return to scale production function $y_j^r = \bar{\psi}^r d_j^r$. Foreign firms incur trade costs τ when exporting to Germany but face no costs when exporting within the RoW.

Workers: Each country o is endowed with an exogenous mass of workers N_o born in that country. Each worker i from o draws an ability vector $\eta_{i,\ell,k}^o$ that is specific to sector k and location ℓ from a Fréchet distribution with shape parameter $\nu > 1$ and scale parameter $A_{o,k,l}$. The scale parameters are assumed to be the same across origins in RoW ($A_{o,k,\ell} = A_{x,k,\ell}$ for $o \in$ RoW), and can be interpreted as the comparative advantage of nationals from RoW and Germans in industry k in ℓ.

Labor supply to sector k in Germany: Workers from o choose the industry k and country ℓ that yield the highest real wage net of iceberg migration costs $\phi_{o,k,\ell}$ with $\phi_{\ell,k,\ell} = 1$. We assume that there is free mobility across countries within RoW and that migration costs to move to ℓ outside RoW are the same (e.g., $\phi_{o,k,\ell} = \phi_{x,k,\ell}$ for $o \in$ RoW). Due to data limitations, we also assume the cost of migration out of Germany is infinity, such that German workers are immobile across countries.[19]

[19]This may not be a strong assumption as the emigration rate of Germans in 2010 is approximately 3.5%, according to the IAB brain-drain dataset.

Following the properties of the Fréchet distribution, the fraction of natives and immigrants from country o who choose industry k in Germany ($\ell = g$) is:

$$\pi_{g,k,g} = \frac{A_{g,k,g}\left(w_{d,k}\right)^{\nu}}{\sum_{k} A_{g,k,g}\left(w_{d,k}\right)^{\nu}} \quad , \quad \pi_{x,k,g} = \frac{A_{x,k,g}\left(\frac{w_{x,k}}{P_g}\right)^{\nu}\phi_{x,k,g}^{-\nu}}{\sum_{k} A_{x,k,g}\left(\frac{w_{x,k}}{P_g}\right)^{\nu}\phi_{x,k,g}^{-\nu} + A_{x,r}\left(\frac{w_r}{P_r}\right)^{\nu}} \tag{11}$$

where w_r and P_r are the wage per efficient unit of labor and price index in RoW. Similar expressions apply for the share of workers from o choosing $\ell \neq g$ (see Appendix **??**). The share $\pi_{x,k,g}$ shows that reducing costs to migrate to Germany increases the supply of immigrants into the country.[20]

Equilibrium and Market Clearing: The equilibrium are prices, efficient wages, and labor allocations such that: workers choose the industry and destination country to work for to maximize income, consumers choose how much of each variety to purchase to maximize utility, firms choose the sourcing strategy and export status to maximize profits, all markets clear, and trade is balanced.

4.2 Firm Heterogeneity and Welfare Gains

This section shows that ignoring heterogeneity in the immigrant share across firms may lead to biased estimates of the welfare gains of immigration. To that end, we compare the analytical welfare gains of a model with heterogeneous immigrant share across firms within a sector ("heterogeneous model") and a model that ignores this heterogeneity but still allows for heterogeneity in innate productivity ("homogeneous model"). The heterogeneous model is a simplified version of our fully quantitative model, where the economy has only one sector and all firms hire some amount of immigrants (e.g., $f_k^{imm} = 0$). The homogeneous model can be a special case of this heterogeneous model with $f_j = 0$ or any model following the Arkolakis et al. (2012) framework. This section assumes that the economy is in trade autarky.[21]

[20] The labor market in Germany is more rigid than other countries (D'Amuri et al., 2010). Such rigidity is not explicitly incorporated into our model since we allow natives to move freely across firms.

[21] Appendix **??** discusses an alternative model where heterogeneity s_j arise from home-bias technological parameter $\beta_{k,j}$.

The native workers' welfare effects of an exogenous change in the number of immigrants, denoted by $\tilde{\mathbb{W}}^{heterog}$, is given by the following change in real wages:

$$\tilde{\mathbb{W}}^{heterog} = -\sum_{j\in J}\frac{\omega_j \tilde{s}_j}{\epsilon-1} = -\frac{\tilde{S}}{(1-\pi)\,\epsilon+\pi\,\sigma-1} = -\underbrace{\frac{\tilde{S}}{\epsilon-1}}_{\tilde{\mathbb{W}}^{homog}}\underbrace{\frac{1}{1+\frac{\sigma-\epsilon}{\epsilon-1}\,\pi}}_{Bias} \tag{12}$$

$$1-\pi \equiv \frac{\sum_{j\in J}\omega_j^S \alpha_j}{\sum_{j\in J}\omega_j \alpha_j} \in (0,1]$$

where we denote $\tilde{x} \equiv dlog(x)$, $s_j \equiv s(\psi_j, f_j)$, S is the native share in the total wage bill in the economy, $\omega_j \equiv \omega(\psi_j, s_j)$ is the share of firm j in expenditure, $\omega_j^S \equiv \omega^S(\psi_j, s_j) = \omega_j \frac{s_j}{\sum_{j\in J}\omega_j s_j}$ is the share of firm j in native employment and $\alpha_j \equiv \alpha(s_j)$ is a strictly decreasing function of s_j that determines the change of s_j in response to a change in the immigrant-to-native wages. The weight π depends on the heterogeneity in immigrant share and it is zero if $s_j = s$ (it occurs when $f_i \to 0$ because $\alpha_j = \alpha$).[22]

Immigration affects the welfare of native workers through two channels: a labor market competition effect and a price effect. The competition effect depends on how substitutable immigrants and natives are at the workplace, as well as how much firms expand due to the higher availability of immigrant labor. The price effect arises because immigration reduces firms' marginal costs, leading to lower prices for goods and services, which increases the purchasing power of workers' wages.

When firms employ the same immigrant share, $\pi = 0$ and the welfare gains on the right-hand side of expression 12 are given by the first component. In these models, the competition effect is characterized by an increase in immigrant shares and a drop in the marginal cost of production of the same proportion for all firms. As firms produce at lower cost and charge lower prices, the purchasing power of wages increases. The native-born real wage increase depends on the size of the inflow and the elasticity of substitution between immigrants and natives in the labor market, which is fully determined by ϵ when

[22] A home-bias model provides a intuitive expression: π is proportional to the weighted variance of s_j ($\pi \propto \sum_j \omega_j (s_j - S)^2$), thus increasing with the heterogeneity in s_j (Oberfield and Raval, 2014).

firms employ the same immigrant share.[23] Note that in the homogeneous model, there is no reallocation of resources across firms because they have the same cost shares.

The welfare predictions of the homogeneous model may be biased for two reasons: firms exhibit different elasticity of marginal cost and elasticity of immigrant-to-native wage bill (substitution elasticity) in response to relative wages, as shown by equations 13 and 14:

$$\tilde{u}_j = (1 - s_j)\,\tilde{W}_{x,j} = (1 - s_j)\,(1 + \zeta(s_j))\,\tilde{w}_x \tag{13}$$

$$\widetilde{(1 - s_j)} - \tilde{s}_j = -(\epsilon - 1)\,(1 + \zeta(s_j))\,\tilde{w}_x \tag{14}$$

where we normalize $w_d = 1$, $\widetilde{(1 - s_j)} - \tilde{s}_j$ is the change in relative wage bill of immigrants relative to natives at firm j, $(\tilde{W}_{x,j} + \tilde{x}_j) - (\tilde{w}_d + \tilde{d}_j)$, $1 + \zeta(s_j) \equiv \frac{\tilde{W}_{x,j}}{\tilde{w}_x} = 1 + \frac{\tilde{n}_j}{\tilde{w}_x}$ is the elasticity of firm j's immigrant bundle cost to the immigrant market wage.

To gradually build intuition on how these mechanisms affect the aggregate welfare, consider first a case where the extensive margin is not active ($\zeta(s_j) = 0$), so the heterogeneity in s_j leads to heterogeneity in the elasticity of marginal cost u_j to w_x but it does not lead to heterogeneity in the substitution elasticity.[24] In this case, the elasticity of substitution between immigrants and natives in the labor market, which we will refer to as the *aggregate* elasticity, equals $(1 - \pi)\,\epsilon + \pi\,\sigma$. The parameter ϵ measures substitution within firms, while σ captures the scale-of-production effect, namely substitution across varieties produced by firms with different immigrant intensities and resulting price declines. The difference between ϵ and σ pin down the direction of the reallocation of natives across firms with different immigrant shares.[25]

In the edge case of $\epsilon = \sigma$, the substitution and scale effects cancel out, and immigrants do not crowd-in or crowd-out native workers. Given that the reallocation of natives across firms is muted, the demand response for native labor and welfare gains is the same as

[23] The more substitutable immigrants and natives are, the lower the productivity gains for firms and the welfare gains for natives. When $\epsilon \to \infty$, the gains are zero because w_d drops in the same proportion as w_x and so does P. If ϵ is finite, $\tilde{w}_x < \tilde{P} < \tilde{w}_d$.

[24] This occurs, for example, in the home-bias model where heterogeneity in immigrant shares comes from heterogeneous $\beta_{k,j}$ as opposed to fixed costs (see Appendix **??** for details).

[25] Mathematically, $\tilde{d}_j - \tilde{d}_{j'} = \frac{\epsilon - \sigma}{\epsilon - 1}\left(\tilde{s}_{dj} - \tilde{s}_{dj'}\right) \approx (\epsilon - \sigma)(s_{j'} - s_j)\tilde{w}_x$

those predicted by the homogeneous model. When the elasticity of substitution within the firm is stronger than the elasticity of demand ($\epsilon > \sigma$), immigrants crowd out natives from immigrant-intensive firms, and natives are reallocated toward native-intensive firms. Such an increase in specialization between natives and immigrants in producing different varieties makes them less substitutable in the labor market than when natives do not reallocate across firms. Given that this reallocation adjustment is absent if firms employ the same immigrant share, the increase in the aggregate demand for natives and welfare is larger in the heterogeneous world. Similarly, when $\epsilon < \sigma$, welfare gains are lower in the heterogeneous world.

Now, consider the case where firms with different immigrant shares adjust the number of sourcing countries differently. The increase in profits following the drop in w_x induces firms to pay more times the fixed cost to source additional immigrants from more origins, which further reduces the cost of production. Therefore, p_j is more elastic to w_x and so is the price index. The resulting decline in the price index is larger when cost reductions are concentrated among large firms, which have a disproportionate weight in consumers' consumption baskets.

To sum up, equation 12 shows that the sign of the bias depends on the sign of $\epsilon - \sigma$, and its magnitude also depends on the correlation between firm size and immigrant share. Given this insight, we will formally test whether $\epsilon - \sigma > 0$, and calibrate the model to match the observed correlation between immigrant share and firm size. We will also validate the quantitative model based on the insights from equation 13 and 14, which highlights the role of the heterogeneous response of u_j (or equivalently sales) and immigrant-to-native wage bill ratio, respectively.

We conclude this section by briefly discussing the implications of our results for the relationship between micro and macro elasticities, and their consequences for welfare. When firm heterogeneity in factor shares does not lead to firm heterogeneity in elasticity of substitution between factors of production, the weighted average in equation 12 coincides with the aggregate elasticity, making the aggregate elasticity sufficient for inferring welfare

gains. However, in our fixed-cost model, where factor share heterogeneity also leads to heterogeneous substitution elasticities, the two diverge. As a result, the aggregate elasticity is not informative about welfare effects because it does not fully capture the first-order impact of price changes. Micro elasticities and firms' shares are needed for welfare analysis.[26]

4.3 Heterogeneity in immigrant share and international trade

We now ask what this firm heterogeneity in immigrant shares implies for the characteristics of countries that make immigrant flows more valuable. In particular, motivated by fundamental theorems of international trade (Samuelson (1948), Rybczynski (1955)), we study if the gains from immigration are larger or lower when a country is open to trade *due to migrant share heterogeneity.*

We compare the welfare gains from immigration in an economy open to trade and the corresponding one under trade autarky. To simplify the analytical expressions, we focus on an open economy where trade is balanced, $\sigma_d = \sigma_x = \sigma$, and firms must pay a fixed cost to export and iceberg trade costs. The welfare gains are given by equation 15:

$$\tilde{\mathbb{W}}^{heterog,open} = -\lambda \frac{\tilde{S}}{(1-\pi^{open})\epsilon + \pi^{open}\sigma - 1} - (1-\lambda)(\tilde{w}_r - \tilde{w}_d) \tag{15}$$

where λ is the aggregate domestic trade share, and π^{open} is defined as before but evaluated in the equilibrium of the open economy. The welfare effects under trade autarky in equation 12 are a special case of equation 15 where $\lambda = 1$ and $\pi(\{s_j, \omega_j\})$ is evaluated in the equilibrium under trade autarky. Throughout this section, we relabel $\tilde{\mathbb{W}}^{heterog}$, $\tilde{\mathbb{W}}^{homog}$, and π in equation 12 as $\tilde{\mathbb{W}}^{heterog,closed}$, $\tilde{\mathbb{W}}^{homog,closed}$, and π^{closed} to distinguish them from their open-economy counterparts.

The analytical expression reveals two differences in the welfare effects of immigration between an open and a closed economy. First, the second term on the right-hand side vanishes in the closed economy because λ equals one. When workers reallocate from the

[26] Mathematically, $\tilde{\mathbb{W}}^{heterog} = -\frac{\tilde{S}}{\epsilon^{agg}-1}\left(\sum_j \omega_j \frac{1-s_j}{1-S}\zeta_j\right)$ where ϵ^{agg} is the aggregate elasticity.

RoW to Germany, labor costs abroad increase relative to native-born wages in Germany ($\tilde{w}_r - \tilde{w}_d > 0$). In an open economy, this reduces the purchasing power of Germans by increasing the price of imported goods, an effect that does not occur under trade autarky. This terms-of-trade-type effect operates even when firms have the same immigrant share and is present in standard quantitative models of international trade and migration.

Second, the value of π in the open economy can be larger or smaller than under trade autarky. This effect operates only if firms differ in the immigrant share; otherwise $\pi = 0$ in an open and closed economy. This is the novel mechanism affecting the role of trade in the wage effects of immigration. Intuitively, the strength of the reallocation of natives toward native intensive firms can be weaker or stronger in an open economy than in a closed economy. To see this, consider two firms: firm j, which is relatively productive and exports when the country is open to trade, and firm j', which is relatively unproductive and does not export. Because more productive firms are more immigrant-intensive, an inflow of immigrants leads firm j to expand production relative to firm j'; more so when the economy is open to trade due to the larger market size for exporters. On the one hand, this stronger increase in the scale of production at the immigrant-intensive firms in an open economy leads immigrants at that firm to displace fewer native coworkers, and reallocation toward native-intensive firms is weaker. On the other hand, the associated stronger increase in profit in the immigrant-intensive firms is larger, so firms source immigrants from more countries, replacing more natives at the firm, and strengthening the reallocation toward native-intensive firms.

According to equation 15, the difference between the welfare gains in an economy with homogeneous immigrant share ($\pi = 0$), with and without trade, is informative about the first mechanism:

$$\tilde{\mathbb{W}}^{homog,open} - \tilde{\mathbb{W}}^{homog,closed} = (1-\lambda)\,\frac{\tilde{S}}{\epsilon - 1} - (1-\lambda)\,(\tilde{w}_r - \tilde{w}_d)$$

The extra difference between the gains in an open relative to a closed economy when firms

employ different immigrant shares is informative about the second mechanism:

$$\left(\tilde{\mathbb{W}}^{heterog,open} - \tilde{\mathbb{W}}^{heterog,closed}\right) - \left(\tilde{\mathbb{W}}^{homog,open} - \tilde{\mathbb{W}}^{homog,closed}\right) = \frac{\tilde{S}}{\epsilon - 1}(\sigma - \epsilon)\left(\lambda \frac{\pi^{open}}{(1-\pi^{open})\epsilon + \pi^{open}\sigma - 1} - \frac{\pi^{closed}}{(1-\pi^{closed})\epsilon + \pi^{closed}\sigma - 1}\right) \tag{16}$$

This expression shows how firm heterogeneity in immigrant share matters for the role of international trade in the gains from immigration. In Section 6.3, we quantify its quantitative relevance by computing the left-hand side.

5 Model Estimation and Validation

We calibrate the parameters of the model in two steps. First, we estimate outside the model the elasticity of demand in Germany and the rest of the world (σ_d and σ_x), the elasticity of substitution between immigrants and natives (ϵ), the elasticity of substitution across origin countries (κ), and the elasticity of labor supply (ν). Second, we calibrate the remaining parameters to match relevant aspects of the data. Finally, Section 5.5 verifies that the calibrated model properly captures the mechanisms through which firm heterogeneity in immigrant share affects welfare by comparing predicted model responses with those observed in the data.

5.1 Elasticity of Demand

Following Oberfield and Raval (2014), we infer the demand elasticity σ_d from firms' markups, i.e., the ratio of revenue to total costs. According to the model, the following moment condition holds:

$$\mathbb{E}\left(\frac{Revenue_j}{Cost_j}\right) = \frac{\sigma_d}{\sigma_d - 1} \tag{17}$$

where j are firms in the non-tradable sector. We compute the average revenue-to-cost ratio across firms and use equation 17 to infer the value of σ_d.[27] The average markup for

[27] Rather than using variable costs in the calibration of σ_d, which is unobservable, we use total costs (the sum of wage bill and material bill), which include fixed costs. As a result, our estimate could be upward-biased. Note that the direction of the bias means that it is even more likely that $\sigma < \epsilon$. Quantitatively, we expect its effect to be small, as the wage bill of occupations related to fixed costs

firms in the non-tradable sector is 1.47, which implies that $\hat{\sigma}_d = 3.1$.

Based on the analogous model's moment condition for firms in the tradable sector, we calibrate the demand elasticity from the RoW σ_x, to match the average revenue-to-cost ratio of firms in the tradable sector, given the calibrated value of σ_d and the average share of exports in their sales.[28] We find that $\hat{\sigma}_x = 4.3$, which implies an average elasticity in the tradable sector of 3.44.

The calibrated values are consistent with previous estimates in the literature, which vary by the granularity of sectoral definitions. For instance, for narrower categories than ours, Broda and Weinstein (2006) found an average elasticity of 4, and for broader categories, estimates tend to be around 0.5 (Cravino and Sotelo (2019), Comin et al. (2021)).

5.2 Elasticity of Substitution Between Natives and Immigrants

We use the first-order conditions with respect to d_j and x_j to obtain equation 18.

$$Log\left(\frac{\text{Wage Bill Immig}_{j,t}}{\text{Wage Bill Natives}_{j,t}}\right) = \frac{\epsilon - 1}{\epsilon} Log\left(\frac{\hat{x}_{j,t}}{N_{g,j,t}}\right) + FE_j + FE_{m,t} + FE_{k,t} + u_{j,t} \quad (18)$$

where $N_{g,j,t}$ is the number of natives employed at the firm at year t and $\hat{x}_{j,t}$ is the immigrant effective units CES composite. We incorporate a rich set of fixed effects FE_j, $FE_{m,t}$, and $FE_{k,t}$ which are firm, commuting zone-year and industry-year fixed effects, respectively. We denote the error term $u_{j,t}$.

In Appendix **??**, we explain in detail how we go from the first order conditions to estimating equation 18. A key part of the process is that we use the Fréchet assumptions for the ability draws to separate ability units from labor units. We then follow the approach by Ottaviano and Peri (2012) to construct the CES composite of immigrant effective units. We use a bootstrap procedure to generate confidence intervals that properly account for the fact that $\hat{x}_{j,t}$ arose from a previous estimation procedure.

accounts for approximately 1% on average of our variable costs.

[28] We use the following condition $\mathbb{E}\left(\frac{Cost_j}{Rev_j}\right) = \frac{Rev_d}{Rev}\frac{Cost_d}{Rev_d} + \frac{Rev_x}{Rev}\frac{Cost_x}{Rev_x} = \frac{Rev_d}{Rev}\frac{\sigma_d - 1}{\sigma_d} + \frac{Rev_x}{Rev}\frac{\sigma_x - 1}{\sigma_x}$

There are two concerns to identify $\frac{\epsilon-1}{\epsilon}$ by estimating equation 18 via ordinary least squares approach (OLS). First, measurement error in $x_{j,t}$ may correlate with firm, industry, or labor market factors determining hiring decisions. Second, omitted variable bias would arise if unobserved shocks to a firm's demand, technology, or shocks to a firm's labor market induce firms to expand and hire more immigrants *relative* to natives.

To address concerns not captured by the fixed effects above, we use an instrumental variables approach. We construct our instrument, denoted by $Z_{j,t}$, following the model's definition of $dLog(\frac{x_{j,t}}{N_{g,j,t}})$:[29]

$$Z_{j,t} = \sum_{o \neq g} \left(\frac{\text{Wage Bill Imm}_{o,j,2003}}{\text{Wage Bill Imm}_{j,2003}} \right) \frac{\Xi_{o,t}}{\Xi_{g,t}} \tag{19}$$

where $\Xi_{o,t}$ is the national employment growth from country group o between 2003 and t. This is a shift-share instrument where the shift is given by the growth of immigrants from nationality o relative to the growth of native workers, and the share is given by the fraction of immigrants of each firm that is accounted for by nationals from o.

Our 2SLS estimate is 0.87, and the F-stat of the first stage is 78.7. This result implies an estimate of ϵ is 7.9 (CI between 4.35 and 29.24).[30] Appendix Section **??** includes the OLS and 2SLS results, and Appendix Section **??** includes pre-trend tests and diagnostics on the validity of our shift-share design (Goldsmith-Pinkham et al., 2020).

5.3 Test for the sign of $\epsilon - \sigma$

Section 4.2 explains that the welfare gains predicted by the homogeneous model are downward biased if $\epsilon > \sigma$ and upward biased if $\epsilon < \sigma$. Therefore, we formally test the null hypothesis that $\epsilon - \sigma < 0$. We use 5,000 bootstrapped samples to obtain the empirical

[29] i.e., $dLog\left(\frac{\hat{x}_{j,t}}{N_{d,j,t}}\right) = \sum_{o \neq g} \left(\frac{\text{Wage Bill Imm}_{o,j,t}}{\text{Wage Bill Imm}_{j,t}}\right) dLog\left(\frac{N_{o,t}}{N_{g,t}}\right)$, where $N_{c,t} \equiv \sum_{j} N_{c,j,t}$.

[30] The elasticity of substitution among workers using a CES aggregator has been estimated in various studies, with notable papers reporting values ranging from 4 to 20 (Burstein et al., 2020; Cortes, 2008; Ottaviano and Peri, 2012). While our estimate is within the range of this literature, differences in the level at which production is defined, the nesting order, and the composition of the immigration shock make direct comparisons challenging. Mahajan et al. (2025) estimate the elasticity of substitution between college-educated immigrants and natives within firms in the U.S. to be 4.3.

distribution for the statistic $\widehat{\epsilon - \sigma}$. We find that $\widehat{\epsilon - \sigma} < 0$ in only 0.007% of the 5000 cases. That is, the p-value of the null hypothesis that $\epsilon - \sigma < 0$ is 0.007, suggesting that the welfare gains of native workers predicted by the homogeneous model might be downward biased. Appendix **??** provides details and the empirical distribution.

5.4 Additional Parameters

This section briefly discusses the calibration of the remaining structural parameters. See Appendix **??** for more details.

The elasticity of labor supply ν is estimated to be 6.14 using the residual variance of wages following the approach used by Hsieh et al. (2019) and Fan (2019). The Cobb-Douglas parameter α is set to 0.68 to match the share of the tradable expenditure of German consumers using World Input-Output Tables (Timmer et al., 2015).

Indirect inference approach: The remaining fourteen parameters are jointly calibrated by minimizing the distance between fourteen model-simulated moments and their empirical counterparts. Table 1 reports the calibrated parameter values, and Appendix Table **??** compares the moments in the data and the model.

Although all parameters are estimated simultaneously, each parameter is primarily informed by a subset of moments. Moments of the joint distribution of firm size and its immigrant share are informative about the parameters of the joint distribution of fixed costs f_j and productivity ψ_j. In particular, the variance of log revenues conditional on the immigrant share in sector k (and on exporter status in the tradable sector) is used to identify the dispersion of productivities $\sigma_{\psi,k}$. The standard deviation of the log immigrant-to-native wage bill ratio in sector k identifies the variability of fixed costs $\sigma_{f,k}$. The difference in the mean of $1 - s_j$ between firms in percentile 90 relative to percentile 50 is used to identify the correlation between productivities and fixed costs $\sigma_{\psi,f,k}$.[31]

[31] Appendix Figure **??** shows that the correlation is much weaker in the non-tradable sector than in the tradable. With $\sigma_{\psi,f,k} = 0$, the model predicts a positive size–immigrant share correlation, but one that is too weak in the tradable sector and too strong in the non-tradable sector. Thus $\sigma_{\psi,f,T}$ must be negative and $\sigma_{\psi,f,NT}$ positive to match the data.

For the remaining parameters, we use the aggregate immigrant share by sector to identify β_k, the distributional share parameter in the production function. The fraction of firms that hire immigrants helps identify the base fixed hiring costs f_k^{imm}. The average immigrant share across all firms and sectors is used to identify ι, the elasticity of $W_{x,j}$ to the mass of countries the firm sources immigrants from. We use the mean ratio of export to domestic revenues for exporters and the fraction of firms that export in the tradable sector to identify the iceberg cost τ and the fixed cost of exporting f_x, respectively. Finally, we use the relative GDP per capita between Germany and the RoW to identify the mean productivity of RoW $\bar{\psi}^r$.

5.5 Model Validation: Heterogeneous Response by Firm Size

Before using the model for counterfactual analysis, we must verify that the calibrated model properly captures the main mechanisms through which immigrant-share heterogeneity affects welfare. Specifically, this heterogeneity in immigrant shares matters because it leads to heterogeneity in the elasticity of unit production costs or, equivalently, sales (equation 13) and the elasticity of the immigrant-to-native wage bill (equation 14) to immigrant labor inflows. Therefore, we validate the model by comparing its predicted changes in firms' sales and the immigrant-to-native wage bill ratio due to an exogenous immigrant inflow with those empirically estimated from the data.

In the model, the change in the outcome variables is only due to the exogenous inflow of immigrants. On the contrary, in the data, firms' outcomes are affected by many factors. We isolate the effects of an exogenous immigration inflow from other factors that are absent in our model by estimating equation 20:

$$y_{j,t} = \theta_1 \, I_{m,t} \; + \theta_2 \, I_{m,t} \, log(emp_{j,2003}) \; + \theta_3 \, X_{j,t} \; + \delta_j \; + \delta_{k,t} \; + \delta_m t \; + u_{j,t} \qquad (20)$$

where $y_{j,t}$ is either log of sales or immigrant-to-native wage bill ratio of establishment j in year t. Sub-indices m and k denote the labor market, defined by commuting zones, and 2-digit industries, respectively. $I_{m,t}$ is the share of immigrants in the total wage bill of the

labor market, and $emp_{j,2003}$ captures the initial establishment size. $X_{j,t}$ is j's initial cost-to-revenue ratio interacted with year dummies, to compare firms with similar mark-ups, and initial employment size interacted with a time trend, which increases the precision of our estimates. In terms of fixed effects, this specification allows for labor markets to be in different linear trends as captured by $\delta_m t$, for factors affecting all establishments in an industry over time captured by $\delta_{k,t}$, and for time-invariant establishment unobservable characteristics δ_j that may confound the effects of immigration. We cluster the error term $(u_{j,t})$ at the industry level and labor market level.

We estimate equation 20 via 2SLS using a shift-share instrument inspired by Card (2001) and Ottaviano et al. (2018). The instrument interacts the labor market share of the wage bill of each origin country in 2003 with a time-shifter that captures the national growth rate, from 2003 to year t, of immigrants from origin o relative to the working-age population growth in Germany. Thus, this shift-share instrument interacts country-specific flows of migration with their initial differential presence in local labor markets in Germany. The interaction term is instrumented by this labor market instrument interacted with the initial size of the firm, $log(emp_{j,2003})$.

This section focuses on the comparison between the data and model responses, and relegates the implementation details and regression tables to Appendix **??**. For the sake of the economic interpretation, we compute the elasticity or semi-elasticity of $y_{j,t}$ to $I_{m,t}$, denoted as $\epsilon^y_{j,t}$, as in equation 21. When y is the log of sales, $\epsilon^y_{j,t}$ equals and elasticity of y, and when y is the wage bill ratio, it equals the semi-elasticity.

$$\epsilon^y_{j,t} \equiv \Big(\hat{\theta}_1 \ + \hat{\theta}_2 \ log(emp_{j,2003})\Big) I_{m,t} \tag{21}$$

Using the calibrated model, we generate a marginal change in the immigrant share in each sector and compute the change in the two outcome variables of interest.[32]

With this model-generated data and the empirical estimates from equation 21, we con-

[32]We lower migration costs such that the total number of immigrants increases by 1% in each sector.

struct Figure 3, treating both datasets in the same way. Specifically, we plot the average effect across all firms within each decile, separately by sector. While the *levels* of the responses between the data and the model are not directly comparable, the *relative* differences across deciles are. The levels are not comparable because the estimates of the empirical regression identify relative effects, but might not identify the level effect of the shock (see Nakamura and Steinsson (2014); Wolf (2023) for a discussion on the "The Missing Intercept Problem"). To compare the relative differences, we normalize both the data and model responses to be zero in the first decile and rescale the normalized values as percentage changes with respect to their own average.[33] Figure 3 shows that, in both sectors, the model-predicted changes closely match the revenue and relative immigrant-to-native wage-bill responses observed in the data. We therefore interpret the model's performance as successful in matching these untargeted moments, which are crucial for the welfare effect of immigration.[34]

6 Aggregate implications

Having validated our calibrated model, we use it to study several counterfactuals. Section 6.1 quantifies the economic and welfare effects of an inflow of immigrants into Germany. Section 6.2 quantifies the importance of accounting for firm heterogeneity for these welfare effects. Section 6.3 studies how openness to trade affects the gains from immigration through firm heterogeneity in migrant shares. Finally, Section 6.4 evaluates the aggregate effects of alternative policies that reduce the fixed costs of hiring for specific firms.

6.1 Aggregate adjustment to an exogenous inflow of immigrants

We use the model to calculate a new equilibrium, where the total number of immigrants in Germany increases exogenously by 20%. The size of the shock mimics the magnitude

[33] We rescale the normalized values by multiplying them by the average response in the data across deciles relative to the average response in the model.

[34] This exercise differs from the EU enlargement analysis in Section 3.3.2, where we isolate a direct reduction in fixed hiring costs. Here, we instead capture a region-specific influx of immigrants keeping the structure of fixed costs unchanged, so that the response of large versus small firms depends on the observed distribution of immigrant shares within sectors.

of the immigration wave that occurred in Germany between 2011 and 2017. According to the OECD, the total number of immigrants in Germany went from 10.55 million in 2011 to 12.74 million in 2017, a 20.7% increase.[35]

We compute the welfare of native-born workers as their real labor income. Table 2 shows that the welfare of native workers would increase by 0.11%, representing $1.8 billion for the aggregate economy.[36][37] The welfare gains of firm owners (1.31%) are larger than those of native workers because they face the same price decreases but do not compete with immigrants in the labor market. Total income, which includes earnings for all workers and profits, increases by 1.28% or $22.5 billion.

Figure 4 shows the reallocation of natives across firms, which illustrates the intuition laid out in Section 4.2. Firms in the top size decile of the tradable sector, being the most immigrant-intensive, absorb a large share of incoming immigrants, leading to the reallocation of many native workers to other firms (approximately 30,000 in total). Given their size, large firms in the non-tradable sector absorb a significant portion of these native workers. This reallocation between firms explains most of the cross-sector reallocation of native workers. Natives also reallocate within sectors toward less immigrant-intensive firms. As previously noted, some small firms in the non-tradable sector with high immigrant shares also displace natives in response to immigration (deciles 3–6), but given their low market shares, the overall effect of this reallocation is limited.

Two forces shape how average native wages adjust across sectors. First, as discussed earlier, immigration affects wages per effective unit of native labor in both sectors, depending on the strength of the substitution and scale effects. Second, a composition effect arises

[35] We reduce the migration cost from the RoW to Germany, $\phi_{x,k,g}$, such that it increases the total stock of immigrants by 20% in each sector. This counterfactual does not aim to evaluate specific post-2011 migration events, such as the EU enlargement or the Syrian refugee crisis.

[36] These effects are relatively large. For reference, estimates of the U.S. welfare gains from China's rise in world trade are of a similar order of magnitude (Hsieh and Ossa (2016), Caliendo et al. (2019))

[37] Monetary gains are expressed in 2019 USD at PPP. Worker gains apply the model's percentage change to OECD average annual wages (per dependent employee, derived from OECD Employment Outlook 2020 by dividing the published per-FTE figure by the ratio of full-time to all-employee weekly hours) multiplied by OECD dependent employees in Germany. Firm-owner gains apply the model's percentage change to aggregate profit income, computed as OECD GDP divided by the model's aggregate markup.

as lower-ability natives reallocate from the tradable to the non-tradable sector. This reallocation lowers the average ability of workers in the non-tradable sector and raises it in the tradable sector. This change in the composition of workers puts additional downward pressure on average native wages relative to the numeraire in the non-tradable sector and partially offsetting the competition effect in the tradable sector. Appendix Table **??** reports changes in wages per efficient unit, wages per native worker, and the price index, all expressed relative to the numeraire (wage in the RoW).[38]

6.2 The Role of Immigrant Share Heterogeneity on Welfare

This section quantifies the importance of the documented heterogeneity in how immigration affects native workers' welfare. As suggested by the analytical results in Section 4.2, one way to do so is by comparing the effects of an exogenous increase in labor supply $\tilde{S}$ across two models: the *heterogeneous model*, our full-fledged baseline model without the restrictions imposed on Section 4.2, and the *homogeneous model*, which does not have within-industry firm heterogeneity in immigrant shares.[39] Note that the homogeneous model allows immigrant shares to differ across sectors and size to differ across firms within industry. Specifically, in terms of primitives, the homogeneous model differs from the heterogeneous model in that fixed costs to hire immigrants are zero ($f_j = f_k^{imm} = 0$), while the innate firm productivity draws ψ_j, β_k and other parameters remain the same. Thus, the comparison across models is informative about the role of within-industry firm heterogeneity in migrant share in the welfare effects of a given immigrant shock $\tilde{S}$.

Given that our comparison model is similar to the closest that has been considered in the literature (di Giovanni et al., 2015), the models' comparison also answers the question "How different are the effects of immigration in a model with within-industry firm heterogeneity in immigrant share relative to the closest model of immigration in the

[38] These findings contrast with Rybczynski (1955)'s theorem, whose assumptions imply that factor prices and factor shares are fixed. Under those conditions, economies absorb cross-country labor reallocation solely through output adjustments: production expands in the immigrant-intensive sector and contracts in the native-intensive one, which requires natives to move from the latter to the former.

[39] Alternatively, one could compare the effects of the same change in the model primitives (e.g., drop in migration costs). This comparison yields an almost identical quantitative conclusion as in Table 2.

literature?"[40]

Table 2 shows that the homogeneous model delivers welfare gains that are 45% of those in the heterogeneous model (0.06pp).[41] This result aligns with the two mechanisms highlighted in Section 4.2 and our calibration. First, given that $\epsilon > \sigma$, immigration increases the specialization of immigrants and natives in producing different varieties, which weakens the competition faced by natives in the labor market and reduces the downward pressure on their wages. Second, the stronger drop in prices in the heterogeneous model follows from the additional decline in unit costs and being concentrated among larger firms, which have a greater weight in consumers' consumption baskets.

These results highlight the importance of accounting for the heterogeneity in immigrant-native hiring decisions across firms within sectors. Immigration triggers a reallocation of native workers across firms, with significant aggregate quantitative implications.

6.3 The Quantitative Role of Trade

Motivated by the discussion in section 4.3, this section studies quantitatively whether the gains from immigration are larger or lower when a country is open to trade *due to firm heterogeneity in immigrant shares*. We do so by quantifying the double-difference stated in equation 16 using our fully-fledged quantitative model. We need to know $\tilde{\mathbb{W}}^{heterog,open}$, $\tilde{\mathbb{W}}^{heterog,closed}$, $\tilde{\mathbb{W}}^{homog,open}$ and $\tilde{\mathbb{W}}^{homog,closed}$. Given that we already computed $\tilde{\mathbb{W}}^{heterog,open}$ and $\tilde{\mathbb{W}}^{homog,open}$, we only need to compute the welfare effects in the economy under autarky. To compute $\tilde{\mathbb{W}}^{heterog,closed}$, we proceed in two steps. First, starting from the baseline equilibrium with heterogeneous firms and trade openness, we eliminate international trade by raising the iceberg trade costs high enough. This resulting new equilibrium serves as the initial equilibrium in this economy under trade autarky. Second, we induce the same $\tilde{S}$ as before and compute the real wage change for native-born workers, $\tilde{\mathbb{W}}^{heterog,closed}$. To compute $\tilde{\mathbb{W}}^{homog,closed}$, we follow the same two-step procedure, but now

[40] In di Giovanni et al. (2015), firms are heterogeneous in their Hicks neutral productivity but homogeneous in their immigrant share (industries are heterogeneous in their factor shares).

[41] The homogeneous model's welfare effects align with Caliendo et al. (2021), who finds that EU labor market integration raised original members' welfare by 0.04% using a model without firm heterogeneity.

the starting point is the equilibrium with homogeneous firms and trade openness.

We find that international trade dampens the gains from immigration, with most of this dampening driven by the differential importance of firm heterogeneity. In particular, welfare gains in an economy with heterogeneous migrant shares under trade autarky are 0.123%, which is 9% (0.010 pp) larger than the 0.113% gains in the baseline economy (e.g., $\tilde{\mathbb{W}}^{heterog,closed}/\tilde{\mathbb{W}}^{heterog,open} = 1.09$).

Part of this dampening arises from mechanisms that operate even when firms within a sector employ identical immigrant shares: $\tilde{\mathbb{W}}^{homog,open}$ and $\tilde{\mathbb{W}}^{homog,closed}$ are 0.051% and 0.055% respectively, implying that terms-of-trade effects reduce the gains from immigration by 0.004 pp (approximately 39% of the total dampening effect). This magnitude is remarkably similar to the findings of Caliendo et al. (2021), who study the effects of the EU Enlargement using a representative-firm model, lending credibility to our results.

The remaining 61% of the total dampening effects (0.006 pp) is due to the heterogeneity in immigrant shares across firms within sectors, as illustrated in equation 16.

These findings indicate that, given current levels of international trade, firm heterogeneity in migrant shares plays a key role in dampening the gains from immigration. Ignoring this heterogeneity may significantly underestimate the role of international trade in mitigating the gains from immigration.

6.4 Comparing policies that affect firm heterogeneity

Beyond policies that affect the inflow of immigrants into the country, which was studied in prior literature, our model allows us to also study policies that alter the hiring costs of immigrants for specific firms. Appendix Section **??** studies alternative policies that reduce immigrant hiring costs for a subset of firms. Policies targeting smaller firms mainly benefit non-tradable sector firms because they face higher hiring frictions.[42] Small firms in the tradable sector are not similarly affected, as the cost reduction does not offset productivity differences with large competitors in the sector. The effects on native workers

[42]Germany implemented the Competence Center for Securing a Skilled Workforce to help small and medium firms hire immigrants (see here for more details). This policy motivated our counterfactual.

are positive but small. We find that lowering fixed costs for high-productivity firms in the tradable sector has the largest welfare effects for natives as firms affected face a more elastic demand, and can expand production without crowding out as many natives.

7 Conclusion

This paper documents substantial heterogeneity in immigrant shares across employers, with larger firms employing immigrants more intensively than smaller firms due to fixed costs of hiring immigrants.

Motivated by these facts, we set up a model to study analytically the welfare implication of firm heterogeneity in immigrant shares for the welfare effects of immigration. First, we show how ignoring heterogeneity in migrant shares may bias welfare predictions. Second, we study the implications of firm heterogeneity in immigrant shares for the characteristics of countries that make immigrant flows more valuable. In particular, we show that, due to migrant-share heterogeneity, international trade can either amplify or dampen the welfare gains from immigration.

We calibrate the model to match our novel evidence and use it to assess the quantitative relevance of these theoretical results. We find that a 20% increase in the number of immigrants raises welfare by 0.11% for native-born workers and by 1.31% for firm owners, corresponding to gains of $1.8 billion and $20.8 billion, respectively. Regarding the first analytical result, we find that a model without within-industry heterogeneity in immigrant shares predicts that the welfare gains for native workers would be approximately 55% lower. This means that immigration induces a reallocation of resources across firms with quantitatively important aggregate implications. Regarding the second analytical result, we find that international trade dampens the welfare gains from immigration, with 61% of this dampening effect due to firm heterogeneity in immigrant share. Thus, ignoring the documented immigrant share heterogeneity may significantly underestimate the role of international trade in mitigating the gains from immigration.

8 Data Availability

Code and information about the proprietary data used in this article can be found in Brinatti and Morales (2026), in the Harvard Dataverse, https://doi.org/10.7910/DVN/ZX3IIY. The authors were granted an exemption to publish the underlying data because access to these data is restricted; the deposit instead includes a simulated/synthetic dataset that enables the code to be run end to end, together with the code for all parts of the paper. The journal verified that the code and simulated data reproduce all tables and figures in the paper, though the simulated data are not designed to reproduce the same numerical results as the restricted data.

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Tables

Table 1: Parameter Calibrated to Match Micro- and Macro- Moments

Parameter description	Parameter	Estimate	Parameter description	Parameter	Estimate
Share of natives, T	β_T	0.89	Covariance of ψ and f_j, NT	$\sigma_{\psi,f,NT}$	30.83
Share of natives, NT	β_{NT}	0.88	Fixed cost of immigrants, T	$f_{imm,T}$	2.79E-05
Dispersion in ψ_j, T	$\sigma_{\psi,T}$	1.15	Fixed cost of immigrants, NT	$f_{imm,NT}$	0.0008
Dispersion in ψ_j, NT	$\sigma_{\psi,NT}$	0.37	Productivity in RoW	ψ_x	1.53
Dispersion in f_j, T	$\sigma_{f,T}$	15207	Fixed cost of exporting	f_g	0.003
Dispersion in f_j, NT	$\sigma_{f,NT}$	16385	Iceberg trade cost	τ	1.35
Covariance of ψ and f_j, T	$\sigma_{\psi,f,T}$	-54.03	Elasticity s_j to n	ι	0.002

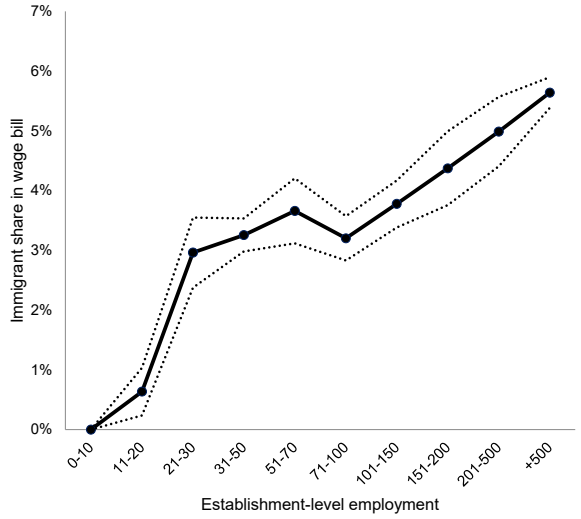
Table 2: Welfare effects with and without firm heterogeneity in the immigrant share

	Open to Trade	Trade Autarky	Autarky/Open
Heterogeneous s_j	0.113%	0.123%	1.09
Homogeneous within-sector s_j	0.051%	0.055%	
Homogeneous/Heterogeneous	0.45		

Notes: For both models, we compute the change in native workers' real wages of going from the observed equilibrium to an equilibrium where the number of immigrants is 20% higher. The heterogeneous model is our baseline model, and the homogeneous model does not have within-industry firm heterogeneity in immigrant shares. The Homogeneous/Heterogeneous row reports the ratio of welfare gains in the homogeneous model relative to the heterogeneous model. The third column reports the relative difference in welfare gains between the economy under trade autarky and the open economy. The bottom row reports the share of the total dampening effect accounted for by within-sector heterogeneity in immigrant shares.

Figures

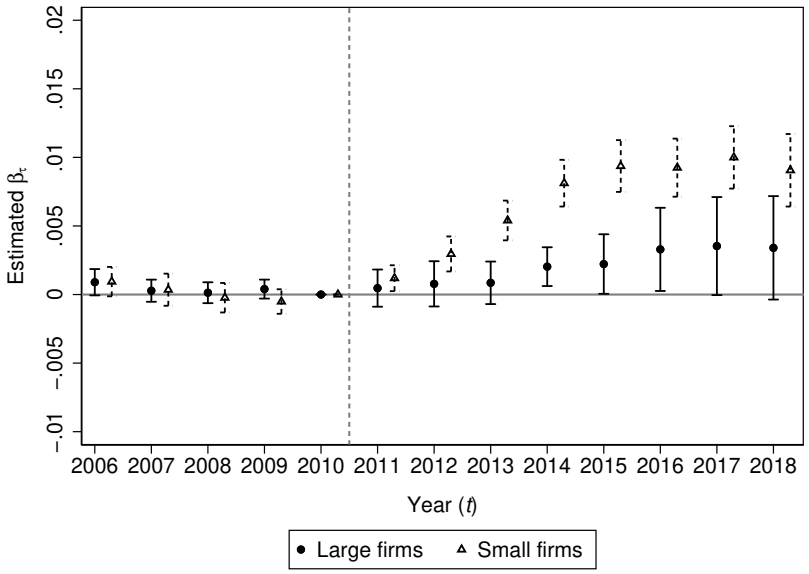
Figure 1: Immigrant share across the establishment size distribution



Notes: We divide establishments into bins based on employment size and, for each bin, plot the median immigrant share of the wage bill. We compute 95% confidence intervals using 200 bootstrap repetitions. We use the years between 2003 to 2011.

Alt text: Line chart showing the median immigrant share of the wage bill (y-axis, 0–7%) by establishment size (x-axis, from 0–10 to 500+ employees, in ten bins). The relationship is increasing overall: the share rises from 0% for the smallest establishments to about 5.6% for establishments with more than 500 employees, with a small dip between the 51–70 and 71–100 employee bins. Dotted lines above and below the main line show 95% confidence intervals from a 200-repetition bootstrap.

Figure 2: Evidence of fixed cost to hire immigrants before 2011

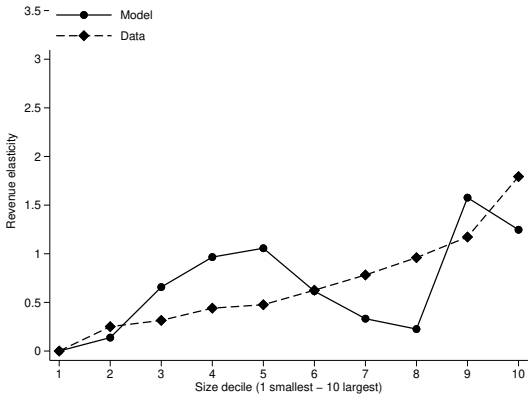


Notes: This figure plots the coefficients β_τ^L and β_τ^{SME} for large and small establishments from equation 1. Large establishments are firms with more than 200 employees. The sample is restricted to all establishments that did not hire NMS or other European migrants in the year 2005 (the year before our sample begins). We cluster standard errors at the establishment level and report 95% confidence intervals.

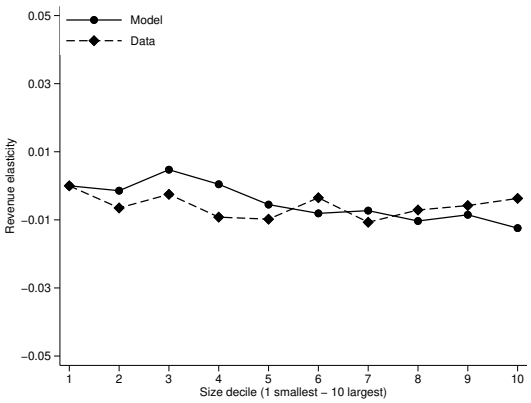
Alt text: Event-study plot showing estimated coefficients for large establishments (more than 200 employees) and small/medium establishments on the hiring of New Member State immigrants relative to a control group, by year before and after the 2011 EU labor-market opening. Coefficients are near zero and similar between the two groups before 2011, then diverge after 2011, with small/medium establishments' coefficients rising above those for large establishments. Vertical bars show 95% confidence intervals from establishment-level clustered standard errors.

Figure 3: Immigration effects by Firm Size: Data vs. Model

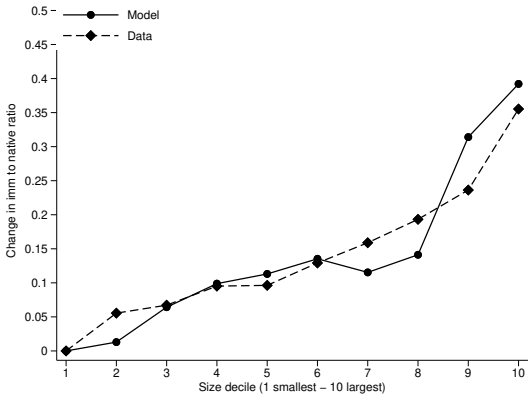
(A) Revenues – Tradable sector



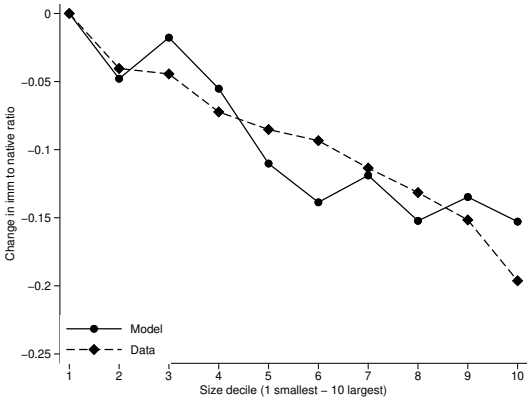
(B) Revenues – Non-Tradable sector



(C) Relative Wage Bill – Tradable sector



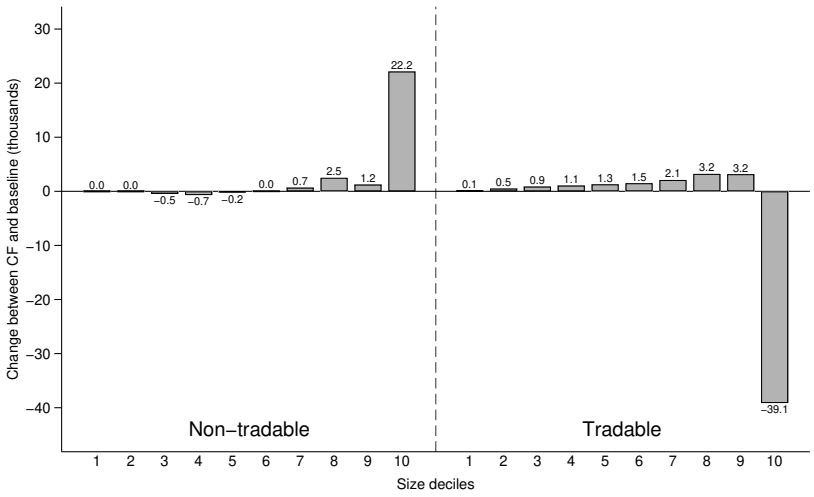
(D) Relative Wage Bill – Non-Tradable sector



Notes: We rank establishments into size-based deciles, with decile 1 being the smallest. We compute the elasticity of revenues (top panels) and immigrant-to-native wage bill ratio (bottom panels) to a 1% increase in the aggregate immigrant share and calculate the mean elasticity across firms in each decile in each sector. We normalize both the data and model responses to be zero in the first decile and rescale the normalized values as percentage changes with respect to their own average.

Alt text: Four-panel line chart comparing model-predicted (solid) and data-estimated (dashed) elasticities across ten firm-size deciles (1 smallest to 10 largest). Panel A, revenue elasticity in the tradable sector: both series rise from 0 to about 1.5–1.8 at decile 10, with model and data tracking closely apart from some model overshoot in the middle deciles. Panel B, revenue elasticity in the non-tradable sector: both series stay close to zero across all deciles. Panel C, change in the immigrant-to-native wage bill ratio in the tradable sector: both series rise steadily from 0 to about 0.35–0.4 at decile 10. Panel D, same ratio in the non-tradable sector: both series decline from 0 to about −0.15 to −0.2 at decile 10. Model and data lines track each other closely in all four panels.

Figure 4: Reallocation of natives across sectors and firms.



Notes: The x-axis shows the distribution of firms in the non-tradable (left) and tradable (right) sectors in terms of deciles of revenues. The bars plot the absolute change in thousands of workers employed in each decile between the baseline and the counterfactual scenario of a 20% increase in the number of immigrants in each sector.

Alt text: Bar chart showing the change in native employment, in thousands, between the baseline and a counterfactual 20% increase in immigrants, by firm-size decile (1–10), separately for non-tradable and tradable sector firms. Most deciles show near-zero change. The largest non-tradable decile shows an increase of 22.2 thousand native workers, while the largest tradable decile shows a decrease of 39.1 thousand, indicating natives reallocating out of large tradable firms and into large non-tradable firms.